Unit: Unit 2: Exponents Radicals
Topic: Operations with Radicals

Name: ____________
Date: __________

Advanced (Habanero)

Q1. Simplify $2\sqrt{3} + 3\sqrt{3}$
(A) $5\sqrt{3}$
(B) $\sqrt{3}$
(C) $6\sqrt{3}$
(D) $2\sqrt{3} + 3$
(E) $2 + 3\sqrt{3}$

Q2. Simplify $\sqrt{12} - \sqrt{3}$
(A) $2\sqrt{3} - \sqrt{3}$
(B) $\sqrt{12} - \sqrt{4}$
(C) $\sqrt{9} - \sqrt{3}$
(D) $\sqrt{15}$
(E) $\sqrt{12} + \sqrt{3}$

Q3. Simplify $(2\sqrt{2})(3\sqrt{2})$
(A) $6\sqrt{4}$
(B) $6\sqrt{2}$
(C) $12\sqrt{2}$
(D) 12
(E) 6

Q4. Simplify $\frac{\sqrt{20}}{\sqrt{5}}$
(A) $2\sqrt{5}$
(B) $\sqrt{20}$
(C) $\sqrt{4}$
(D) 2
(E) $\sqrt{5}$

Q5. Simplify $\sqrt{16} + \sqrt{9}$
(A) $4 + 3$
(B) 5
(C) 7
(D) $\sqrt{16} + \sqrt{9}$
(E) $\sqrt{25}$

Q6. Simplify $(\sqrt{3} + \sqrt{2})^2$
(A) $3 + 2 + 2\sqrt{6}$
(B) $5 + 2\sqrt{6}$
(C) $3 + 2$
(D) $\sqrt{3} + \sqrt{2}$
(E) 5

Q7. Simplify $\frac{\sqrt{32}}{\sqrt{2}}$
(A) $\sqrt{16}$
(B) $\sqrt{4}$
(C) $4\sqrt{2}$
(D) $\sqrt{8}$
(E) 8

Q8. Simplify $\sqrt{8} \cdot \sqrt{2}$
(A) 4
(B) $\sqrt{16}$
(C) $\sqrt{4}$
(D) $2\sqrt{2}$
(E) $\sqrt{2}$

Open Ended Questions (FRQ)

Q9. Simplify $(\sqrt{2} + \sqrt{3})(\sqrt{2} - \sqrt{3})$ and show all steps.

Q10. If $x = \sqrt{5}$ and $y = \sqrt{3}$, simplify $x^2 y$ and show all steps.

Chef's Tips

- Emphasize the importance of simplifying radicals.
- Remind students to check their work by plugging in values.
- Make sure students show all steps in free-response questions.